Project Report

Predictive Pulse – Hypertension Risk Detection using Machine Learning

Date	08 March 2026
Team ID	—
Project Name	Predictive Pulse: Harnessing Machine Learning for Blood Pressure Analysis
Maximum Marks	—

1. Project Title

Predictive Pulse: Harnessing Machine Learning for Blood Pressure Analysis

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2. Project Description

System Overview

Predictive Pulse is a machine learning-based web application designed to predict hypertension risk using patient health parameters and symptoms. The system analyzes user inputs such as age group, blood pressure levels, medication history, and symptoms to determine the stage of hypertension and provide health recommendations.

Problem Domain

Hypertension is a major health concern worldwide and often remains undetected until severe complications occur. Traditional diagnosis requires hospital visits and medical tests. Many people lack access to quick preliminary health assessment tools.

Core Technology

- Machine Learning: Logistic Regression (Scikit-learn)
- Backend: Python Flask
- Frontend: HTML5, CSS3
- Model Storage: Joblib (.pkl model file)

Key Capabilities

- Predicts hypertension stage based on user health inputs
- Displays risk level and confidence score
- Provides medical recommendations and lifestyle advice
- Accessible through a simple web browser interface

Target Users

- Individuals concerned about blood pressure health
- Healthcare students studying medical AI systems
- People interested in preventive healthcare tools

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3. Application Scenarios / Use Cases

Scenario 1 – Health Monitoring

- A user experiencing symptoms such as headache or dizziness enters health information into the system. The application predicts hypertension risk and provides recommendations.

Scenario 2 – Preventive Healthcare

- Health-conscious individuals use the system to check their hypertension risk before visiting a doctor.

Scenario 3 – Educational Use

- Students studying machine learning and healthcare technology use the project to understand how AI models can be integrated into web applications.

Scenario 4 – Healthcare Awareness

- The system can be used in health awareness programs to demonstrate how AI can assist in early disease detection.

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4. Technical Architecture Overview

3-Tier Architecture

Frontend Layer

- HTML/CSS user interface for entering health details
- Displays prediction results and recommendations

Web Server Layer

- Python Flask application handling user requests
- Processes input data and interacts with the machine learning model

Machine Learning Layer

- Logistic Regression model trained on hypertension dataset
- Predicts hypertension stage and confidence score

Communication Flow

User → Web Interface → Flask Server → Data Preprocessing → Machine Learning Model → Prediction Result → Browser

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5. Prerequisites

- Software
- Python 3.8 or above
- pip package manager
- VS Code or any Python IDE
- Web browser

Libraries

- Flask
- NumPy
- Scikit-learn
- Joblib

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6. Prior Knowledge Required

- Basic Python programming
- Machine learning fundamentals
- Flask web framework basics
- HTML and CSS for web interface design
- Understanding of medical health parameters

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7. Project Objectives

Technical Objectives

- Develop a machine learning model to predict hypertension stages
- Integrate the model with a Flask web application
- Provide real-time predictions based on user input

Performance Objectives

- Fast prediction response time
- Accurate risk classification
- User-friendly interface

Learning Objectives

- Understand integration of ML models with web applications
- Explore healthcare applications of machine learning

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8. System Workflow

- User opens the web application in a browser
- User enters health details such as age group, symptoms, and blood pressure
- Flask server receives the form submission
- Input data is encoded and preprocessed
- Machine learning model analyzes the data
- Model predicts hypertension stage
- System generates medical recommendations
- Results are displayed on the web interface

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9. Milestone 1: Requirement Analysis & System Design

Problem Definition : Detect hypertension risk using machine learning and provide instant feedback through a web interface.

Functional Requirements

- User health data input form
- Machine learning prediction system
- Display risk level and recommendations

Non-Functional Requirements

- Fast prediction response
- Simple and intuitive user interface
- Secure handling of user input data

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10. Milestone 2: Environment Setup

Steps:

- Install Python and required libraries
- Create project folder structure
- Train machine learning model using dataset
- Save trained model using Joblib

Example command:

1. python -m venv venv
2. pip install flask numpy scikit-learn joblib

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11. Milestone 3: Core System Development

Feature 1 – Machine Learning Model

- Logistic Regression model trained to classify hypertension stages.

Feature 2 – Flask Web Application

- Flask server processes form data and connects frontend with ML model.

Feature 3 – Web Interface

- HTML and CSS based dashboard for user interaction.

Feature 4 – Recommendation System

- Displays health advice based on predicted risk level.

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12. Milestone 4: Integration & Optimization

- Integration of machine learning model with Flask server
- Implementation of input validation and error handling
- Optimization of prediction speed and UI design

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13. Milestone 5: Testing & Validation

Test ID	Input	Expected Output	Status
TC-01	Normal BP values	Normal risk prediction	PASS
TC-02	Slightly high BP values	Stage-1 hypertension	PASS
TC-03	High BP values	Stage-2 hypertension	PASS
TC-04	Very high BP values	Hypertensive crisis warning	PASS
TC-05	Missing fields	Error message displayed	PASS

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14. Deployment

Platform

- Local development server

Steps

1. Install required libraries
2. Run Flask server :**python app.py**
1. Open browser and visit: **http://localhost:5000**

Future deployment can use cloud platforms such as AWS, Render, or Railway.

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15. Project Structure

```
Predictive_Pulse/
│
├── app.py
├── logreg_model.pkl
├── Predictive_Pulse_model.ipynb
│
├── templates/
│   └── index.html
│
└── static/
    └── style.css
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16. Results

The system successfully predicts hypertension stages based on user input data. The web application provides clear risk levels and medical advice, demonstrating how machine learning can assist in early health risk detection.

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17. Advantages & Limitations

Advantages

- Early detection of hypertension risk
- Easy-to-use web interface
- Fast machine learning predictions
- Supports preventive healthcare awareness

Limitations

- Depends on accuracy of input data
- Not a replacement for professional medical diagnosis
- Limited dataset may affect prediction accuracy

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18. Future Enhancements

- Integration with wearable health devices
- Mobile application version
- Advanced deep learning prediction models
- Cloud deployment for public access
- Integration with hospital healthcare systems

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19. Conclusion

Predictive Pulse demonstrates how machine learning can be applied in healthcare to provide quick hypertension risk assessment. By integrating a logistic regression model with a Flask web application, the system provides users with instant predictions and health recommendations, promoting preventive healthcare and awareness.